Neel Gali

Software Engineer

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Professional Experience

Software Engineer *Jul 2025 - Present*

Epik Solutions | Enterprise FinTech Platform

- Built a **Python** market data ingestion framework handling real-time equity, futures, and options tick feeds; normalized 15+ instrument streams with configurable deduplication and alerting thresholds.
- Built **Python** time series framework using **TimescaleDB** for high-frequency OHLCV storage, delivering sub-10ms query response across 3 years of historical market data for 20+ instruments.
- Engineered **Python** trade ingestion microservices processing 50K+ high-frequency trade events per second over **Kafka**, with idempotent handlers, configurable dead-letter queues, and full audit trail persistence.
- Developed **Python** adapters integrating three external market data vendor APIs with retry logic, circuit breakers, and vendor-specific protocol normalization into the Beacon platform ingestion pipeline.
- Extended the platform application framework with a **Python** plugin system, letting engineers register custom data processors and schedule jobs without modifying core platform configuration files.
- Built **pandas** and **NumPy** financial analytics modules computing rolling VWAP, momentum signals, and volatility metrics, feeding enriched instrument data into the Beacon platform real-time dashboard.
- Implemented **Python** order book reconstruction from Level 2 market data feeds using min-heap priority queue, enabling sub-millisecond best bid/offer spread calculation for 50+ equity instruments.
- Enforced **Pydantic** data contract schemas across all ingestion pipeline message types, automatically rejecting malformed upstream payloads before they reached downstream time series or analytics storage.

Software Engineer *Jul 2022 - Jun 2025*

Samay Systems LLC | B2B Data & Compliance Platform

- Built **Python** ETL pipelines on **AWS Lambda** and **S3** processing 10M+ daily compliance records, reducing data ingestion lag from 4 hours to under 20 minutes.
- Architected **Kafka** event streaming for real-time compliance data sync across six microservices, layering **Redis** caching that cut downstream API latency by 65% at peak load.
- Designed **PostgreSQL** relational schemas and **MongoDB** document collections for a dual-store compliance system, tuning each to its query patterns and cutting ORM overhead by 40%.
- Integrated seven third-party compliance data providers via REST and SFTP adapters in **Python**, building idempotent reconciliation logic that detected and resolved cross-system data discrepancies automatically.
- Tuned **Python** data pipeline throughput using connection pooling, batched writes, and async I/O, sustaining 3x data volume growth with zero additional infrastructure or latency degradation.
- Deployed and maintained **FastAPI** REST services on **Azure** for compliance report delivery, integrating **Azure Blob Storage** to archive processed data exports with tenant-scoped access controls.
- Implemented **PostgreSQL** date-range partitioning on 500M+ compliance event records, cutting query scan costs by 70% and enabling automated partition pruning for regulatory data retention policies.

Software Engineer *Jul 2020 - Aug 2022*

Zemoso Technologies | LabOps Platform

- Developed **Python** REST APIs for a LabOps platform, leveraging priority queues and graph algorithms for sample processing workflows, reducing average API response times by 35%.
- Authored optimized **PostgreSQL** queries and stored procedures for lab sample tracking, reducing report generation time by 30% through strategic index tuning and targeted query denormalization.
- Developed **Java Spring Boot** microservices alongside **Python** services in a multi-language stack, maintaining 75%+ test coverage with JUnit and pytest across all platform service boundaries.
- Containerized **Python** microservices using **Docker** and provisioned cloud environments through **Terraform**, enabling local development setups and reducing new-developer onboarding and configuration time by 60%.
- Refactored **Python** ingestion jobs to async I/O using **asyncio**, reducing per-batch processing time by 55% and supporting concurrent polling across multiple upstream data sources simultaneously.

Skills

Languages: Python, Java, SQL, JavaScript, Bash
Data & Streaming: Kafka, Redis, ETL pipelines, pandas, NumPy, market data ingestion, time series (TimescaleDB, OHLCV), HFT trade ingestion
Databases: PostgreSQL, MongoDB, TimescaleDB
Cloud & Infra: AWS (Lambda, S3), Azure (App Service, Blob Storage), Docker, Terraform
Frameworks & Tools: FastAPI, Pydantic, Spring Boot, asyncio, pytest, JUnit, GitHub Actions

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India